Homework 10

Week 12

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1 Sequential Circuit Testing

Problem 1. Find a checking sequence of this state table using the successor tree.

State Table		
Present State	**Input**	
	$X = 0$	$X = 1$
A	B,1	C,0
B	A,0	D,1
C	B,0	A,0
D	C,1	A,1

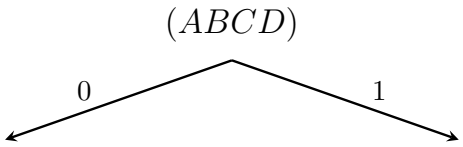
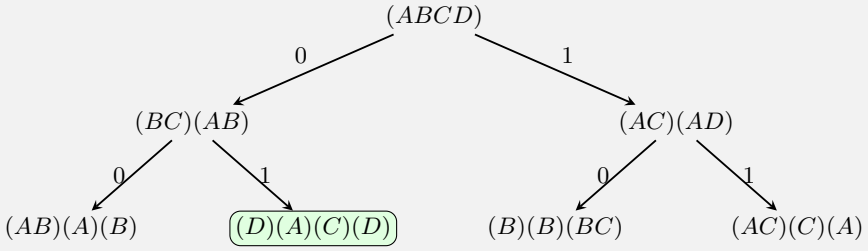


Table 1: State transition and output table

Solution

Successor tree (each node groups the states that are still indistinguishable, by output):



The node reached by 01 contains only singletons, so $X = 01$ is a distinguishing sequence:

Initial state	A	B	C	D
Response to $X = 01$	11	00	01	10
Final state	D	C	D	A

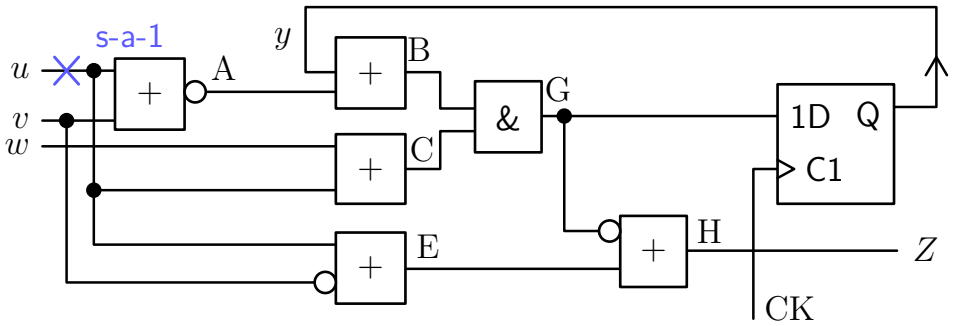
110 is a synchronizing sequence (every state goes to B), so it is used to initialize the machine.
Checking sequence ($X = 01$):

Part	Purpose	Input	Output	End state
Initialize	synchronize to B	110	–	B
Identify states	$B \xrightarrow{X} C \xrightarrow{X} D \xrightarrow{X} A \xrightarrow{X} D \xrightarrow{X} A$	01 01 01 01 01	00 01 10 11 10	A
Check transitions	$A \xrightarrow{0/1} B$	0 01	1 00	C
	$C \xrightarrow{0/0} B$	0 01	0 00	C
	$C \xrightarrow{1/0} A$	1 01	0 11	D
	$D \xrightarrow{0/1} C$	0 01	1 01	D
	$D \xrightarrow{1/1} A$	1 01	1 11	D
	transfer $D \to A$	01	10	A
	$A \xrightarrow{1/0} C$	1 01	0 01	D
	transfer $D \to B$	00	10	B
	$B \xrightarrow{0/0} A$	0 01	0 11	D
	transfer $D \to B$	00	10	B
	$B \xrightarrow{1/1} D$	1 01	1 10	A

Input: 110 0101010101 001 001 101 001 101 01 101 00 001 00 101

Expected output (after initialization): 0001101110 100 000 011 101 111 10 001 10 011 10 110

Problem 2. Find the test sequence for $s-a-1$ on line u.



Solution

Circuit equations (y = flip-flop output, Y = next state):

$$A = \overline{u+v}, \quad B = y + A, \quad C = u + w, \quad E = u + \overline{v}, \quad G = BC, \quad Z = H = \overline{G} + E, \quad Y = G.$$

Faulty machine ($u = 1$): $E = 1 \Rightarrow Z = 1$ for every input and state.

Good machine: we need $Z = 0$, i.e. $u = 0$, $v = 1$, $G = 1$. With $u = 0, v = 1$: $A = 0$, so $G = y\,w$, requiring $y = 1$ and $w = 1$. Since the initial state is unknown, $y = 1$ must be set first: with $u = 0, v = 0$, $A = 1 \Rightarrow G = w$, so $w = 1$ gives $Y = 1$ for any y.

Time	u	v	w	y	A	B	C	E	G	Y	Z	Z_{faulty}
1	0	0	1	x	1	1	1	1	1	1	1	1
2	0	1	1	1	0	1	1	0	1	1	**0**	**1**

$$(u\,v\,w) = 001,\ 011$$

The fault is detected at time 2 ($Z = 0$ good, $Z = 1$ faulty).